

Clinical Voice Agent Architecture

Browser React Demo: microphone, transcript entry, speech synthesis, barge-in

- > FastAPI REST/WebSocket API
- > STT Adapter: browser/local demo transcript, production streaming STT
- > Clinical Agent: language detection, session state, intent, tool planning
- > Tool Registry: list doctors, book, reschedule, cancel, campaign outcome
- > Scheduling Engine: availability, conflict prevention, alternatives
- > Repository Boundary: in-memory demo, Redis TTL sessions, Postgres durable records
- > TTS Adapter: demo audio URL, production streaming voice synthesis

Memory

Session memory stores active intent, pending confirmation, language, transcript.

Patient memory stores language preference, preferred doctor, notes, prior interactions.

Latency

Each turn logs stt_ms, language_ms, reasoning_ms, tts_ms, first_audio_ready_ms.

Target budget is less than 450 ms from speech end or transcript receipt to first audio.